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ANALYSIS OF FOOD INVENTORY MANAGEMENT IN NUTRITION UNIT OF KUMALA SIWI HOSPITAL MIJEN KUDUS

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A B S T R A C T

The hospital unit installation is a unit that handles patient food needs which requires effective and efficient raw material inventory management. The food raw materials analysed are rice, chicken eggs, melons, carrots, watermelon, and catfish. The problem faced by the nutrition installation is the lack of accurate control of raw materials. The purpose of this study was to determine the inventory management of food raw materials according to hospital policy and EOQ calculations. The method used is quantitative descriptive method. The data needed are usage costs, ordering costs, quantity needed, storage costs, order frequency. The analysis method used is the calculation of economic order quantity (EOQ). The results showed a decrease in the frequency of ordering in a year according to EOQ, ordering rice 31 times, chicken eggs 25 times, carrots 27 times, melons 8 times, watermelons 10 times, catfish 21 times. In addition, with the EOQ calculation, the total ordering cost was able to save 82.84% by Rp 2,124,000 from Rp 2,564,000 decreased to Rp 440,000 while the usage cost saved 0.47% by Rp 516,710 from Rp 110,812,200 to Rp 110,295,490.

1. INTRODUCTION

The nutrition unit is a field in the hospital that performs nutrition services with food services for patients carried out every day with a regulated pattern. Therefore, the supply of raw materials for each hospital is very important because food raw materials in hospitals are supporting the improvement of patient nutrition. The existing inventory in each hospital is certainly different, both in terms of quantity and in terms of type. This is because each hospital has a different number of patients.

Food raw materials in hospitals are one of the determining factors in the smooth running of hospital services, so hospitals must have sufficient supplies of raw materials to support service activities. Therefore, the function of food raw material inventory management has an important role and must be owned by every hospital. Each hospital must be able to determine the optimal raw material requirements so that there is no ordering of raw materials in quantities that are too small or too large.

Inventory management is one of the important steps in organising food in the nutrition installation. Insufficient food supplies can cause obstacles in daily operations, while excessive food supplies will also cause costs that burden the nutrition installation, for example the cost of storage and maintenance of unused supplies, another problem is that if the excess material is a material with a short economic life, the raw material will be quickly damaged before use (Firmansyah et al., 2022).

Planning and procurement of food raw materials in the Nutrition Unit of Kumala Siwi Mijen Kudus Hospital is usually carried out by the nutritionist in charge. Ordering food raw materials to suppliers is made every day, the type of raw material ordered is based on the total number of food orders for inpatients that day. In processing food raw materials, depending on the number of inpatients and meal orders for employees by other units on that day.

Based on the initial review that the author has researched, problems that arise in the fulfilment of food raw materials in the Nutrition Unit of Kumala Siwi Mijen Kudus Hospital, namely; the number of inpatients who often change up or down suddenly, affecting the supply of food raw materials, if the number of patients drops suddenly, the raw materials will experience excess stock or a shortage if the opposite happens. In addition, the price of the main raw materials rises drastically, so that the approved procurement costs are insufficient to make payments for the purchase of food raw materials.

Research by Sartika et al (2021) at Besemah Hospital found that there was over stock in the provision of food so that there were many leftovers, such as instant drinks, snacks, seasonings, staple foods. Apart from dry food ingredients, there are wet food ingredients that experience leftovers such as tomato, cabbage, and carrot vegetables, because experiencing the rest sometimes these vegetables decay.

To fulfil the needs of food raw material supplies, of course, good food raw material inventory management is also needed. Inventory management is needed by the hospital by planning to determine the amount of food that must be purchased and paying attention to the timing of purchasing raw materials so that hospital food processing production activities are not disrupted considering that food needs must be maintained continuously. Planning can be done by determining the amount of food that must be purchased, the costs required, and determining the right time to purchase supplies.

The EOQ model is one of the earliest and most widely acknowledged control techniques. A key benefit of the procurement model using the EOQ method is its ability to manage the planning of goods procurement. When proper recording, reporting, and information systems are implemented, it allows for planning that closely aligns with actual needs, leading to minimal inventory and improved availability. Additionally, it helps reduce the amount of working capital needed, with continuous supervision and monitoring of inventory to prevent the risk of overstocking and delays in purchases (Alhamidy, 2017).

Based on the problems that occur above, the authors are interested in knowing how the management of food raw material inventory in the Nutrition Unit of Kumala Siwi Mijen Kudus Hospital with the title ‘Inventory Management of Food Raw Materials Nutrition Unit of Kumalasiwi Mijen Kudus General Hospital (RSU)’.

2. LITERATURE REVIEW

a. Inventory Management

Inventory refers to the quantity of items a company holds, which can be in the form of finished products, semi-finished goods, or raw materials. According to Sofyan Assauri, as cited in the book by Marihot and Dearlina Sinaga (2015), inventory is considered a current asset that includes goods owned by the company for the purpose of being sold during a normal business cycle. It also includes goods that are still in production or raw materials waiting to be used in the production process.

According to Heizer (2017), the function of inventory is to provide a selection of goods in order to meet anticipated consumer demand, and avoid companies from fluctuations in consumer demand.

b. Theory of EOQ

The definition of EOQ according to Arifin (2019) is the amount of raw material purchases at each order at the lowest cost. Meanwhile, Sofyan (2018) defines EOQ as the number or amount of orders held should result in the costs incurred in supplying are minimal.

This approach aims to maintain the lowest possible inventory levels while minimizing costs. By applying the Economic Order Quantity (EOQ) method, a company can reduce the likelihood of stock shortages, ensuring that the production process is not disrupted. Additionally, it helps save on inventory costs by optimizing the management of raw materials

According to Gitosudarmo, (2020) EOQ is actually the most economical volume or amount of purchases to be made at each purchase. To meet these needs, the most economical fulfilment of needs (purchases) can be calculated, namely the number of goods that will be obtained by purchasing using minimal costs.

3. RESEARCH METHOD

This study employs a descriptive quantitative research method, following a quantitative approach. Quantitative research is characterized by being systematic, well-planned, and having a clear structure from the initial stages through the development of the research design.

The variable studied is the efficient supply of food ingredients using the economic order quantity (EOQ) method in the Nutrition Installation of RSU Kumala Siwi Mijen Kudus. The place of this research was carried out at the Nutrition Installation of Kumala Siwi Mijen Kudus Hospital. The time needed to collect data is February 2024.

The writer used three technique of collecting data, those are observation, interview and documentation. Data was analyzed using pattern below:

$$EOQ = \sqrt{\frac{2SD}{H}}$$

Q = economical order quantity

D = the quantity needed in units per year

S = ordering cost for one time order

H = Storage cost of unit per year

4. RESULT AND DISCUSSION

a. The description of ordering and usage cost of food material based on hospital Management

Based on the analysis above, it can be summarised in the table below:

No	Food material	Ordering frecuency (times)	Total order (Kg)	Total usage (Kg)	Price (Rp)	Total usage (Rp)
(1)	*(2)*	*(3)*	*(4)*	*(5)*	*(6)*	*(7=5x6)*
1	Rice	96	2400	2350	Rp12,108	Rp28,488,000
2	Eggs	120	980	980	Rp29,467	Rp28,876,700
3	Carrots	180	1344	1344	Rp12,575	Rp16,900,000
4	Melon	180	388	388	Rp15,083	Rp5,854,500
5	Water melon	180	528	528	Rp12,583	Rp6,647,000
6	Catfishes	76	864	864	Rp27,833	Rp24,046,000

From the data above, it is known that every year the hospital orders rice 96 times, used 2359 kg at an average price of Rp 12,108. Ordering chicken eggs 120 times, used 980 kg at an average price of Rp 29,467. Ordered carrots 180 times, used 1344 kg at an average price of Rp 12,575. 180 orders of melon, used 388 kg at an average price of Rp 15,083. Ordering watermelon 180 times, used 528 kg at an average price of Rp 12,583. And ordering catfish 76 times, used 864 kg at an average price of Rp 27,833.

b. Ordering cost

Ordering cost is the cost that

management needs to prepare in purchasing and ordering goods. Ordering costs can include administrative costs, shipping costs, and other costs. The cost of ordering food raw materials incurred by the kumala siwi kudus hospital in 2023 can be seen in the following table:

No	Food material	Communication	Transportation	Order frequency	Total Communication cost	Total Transportation cost	Total ordering cost
(1)	*(2)*	*(3)*	*(4)*	*(5)*	*(6=3x5)*	*(7=4x5)*	*(8=6+7)*
1	Rice	Rp500	Rp5,000	96	Rp48,000	Rp480,000	Rp528,000
2	Eggs	Rp500	Rp3,000	120	Rp60,000	Rp360,000	Rp420,000
3	Carrots	Rp500	Rp2,000	180	Rp90,000	Rp360,000	Rp450,000
4	Melon	Rp500	Rp2,000	180	Rp90,000	Rp360,000	Rp450,000
5	Water melon	Rp500	Rp2,000	180	Rp90,000	Rp360,000	Rp450,000
6	Catfishes	Rp500	Rp3,000	76	Rp38,000	Rp228,000	Rp266,000

Based on the table above, it can be seen that the ordering costs incurred by the nutrition installation of the kumala siwi kudus hospital to order food raw materials in the form of rice every month are Rp 44,000 so that the total ordering cost for a year in 2023 is Rp 528,000, chicken egg raw materials every month are Rp 35,000 so that the total ordering cost for a year is Rp 420,000, ordering carrot raw materials every month is Rp 37, 500 so that the total cost of ordering for a year is Rp 450,000, ordering melon raw materials every month is Rp 37,500 so that the total cost of ordering for a year is Rp 450,000, ordering watermelon raw materials every month is Rp 37,500 so that the total cost of ordering for a year is Rp 450,000, and ordering catfish raw materials every month is Rp 22,000 so that the total cost of ordering for a year is Rp 266,000.

c. Holding cost

Holding costs refer to the expenses incurred for storing or maintaining inventory over a specific period. These costs also encompass the operational expenses required for warehouse operations.The storage costs that researchers can take at the nutrition installation of the Kumala Siwi Kudus Hospital are the costs of electricity, salaries and water maintenance.

From the analysis, it is known that the percentage of storage costs for rice raw materials is 36.42%, chicken eggs are 15.19%, carrots are 20.83%, melons are 6.01%, watermelons are 8.18%, catfish are 13.39%.

d. Ordering and usage cost based on EOQ analysis

The calculation of the optimal order quantity for food raw materials in 2023 using the Economic Order Quantity (EOQ) method at the Kumala Siwi Kudus Hospital's nutrition installation requires certain data on the hospital's food raw material supply for that year. The necessary data includes the annual demand for food raw materials (D), the cost per order (S), and the holding cost per unit of food ingredients (H), as shown in the table below:

Based on the analysis, there is a decrease in the frequency of ordering rice to 31 times in one year and 77 kg each time. Ordering chicken eggs to 25 times a year and 39 kg each time. Decrease in carrot ordering frequency to 27 times a year and 51 kg per order. Decrease in ordering frequency of melon to 8 times in one year and 46 kg each order. Decrease in ordering frequency of watermelon to 10 times in one year and 51 kg each order. Decrease in ordering frequency of catfish to 21 times in one year and 40 kg each order.

No	Food material	Total order (Kg)	Ordering frecuency (times)	Price	Ordering cost	Total usage cost	Total ordering cost
(1)	*(2)*	*(3)*	*(4)*	*(5)*	*(6)*	*(7=3x4x5)*	*(8=4x6)*
1	Rice	77	31	Rp12,108	Rp5,500	Rp28,901,796	Rp170,500
2	Eggs	39	25	Rp29,467	Rp3,500	Rp28,730,325	Rp87,500
3	Carrots	51	27	Rp12,575	Rp2,500	Rp17,315,775	Rp67,500
4	Melon	46	8	Rp15,083	Rp2,500	Rp5,550,544	Rp20,000
5	Water melon	51	10	Rp12,583	Rp2,500	Rp6,417,330	Rp25,000
6	Catfishes	40	21	Rp27,833	Rp3,500	Rp23,379,720	Rp73,500

The table above shows that the amount of rice usage is Rp 28,901,796 in a year and the total order cost is Rp 170,500, the amount of chicken egg usage is Rp 28,730,325 in a year and the total order cost is RP 87,500, the amount of carrot usage is Rp 17,315,775 in a year and the total order cost is Rp 67, 500, the amount of melon usage is Rp 5,550,544 in a year and the total order cost is Rp 20,000, the amount of watermelon usage is Rp 6,417,330 in a year and the total order cost is IDR 25,000, and the amount of catfish usage is Rp 23,379,720 in a year and the total order cost is Rp 73,500.

e. The inventory management before and after EOQ analysis

Inventory control of food raw materials according to conventional methods applied by hospitals can actually be calculated by the EOQ method. By knowing the results of the comparison, the company can find out which method is optimal and able to generate the minimum cost. The comparison of inventory control before and after the EOQ calculation can be seen in the following table:

No	Food material	Before EOQ analysis		After EOQ Analysis	
		Total usage cost	Total ordering cost	Total usage cost	Total ordering cost
(1)	*(2)*	*(3)*	*(4)*	*(5)*	*(6)*
1	Rice	Rp28,488,000	Rp528,000	Rp28,901,796	Rp170,500
2	Eggs	Rp28,876,700	Rp420,000	Rp28,730,325	Rp87,500
3	Carrots	Rp16,900,000	Rp450,000	Rp17,315,775	Rp67,500
4	Melon	Rp5,854,500	Rp450,000	Rp5,550,544	Rp20,000
5	Water melon	Rp6,647,000	Rp450,000	Rp6,417,330	Rp25,000
6	Catfishes	Rp24,046,000	Rp266,000	Rp23,379,720	Rp73,500
Total		**Rp 110,812,200**	**Rp 2,564,000**	**Rp 110,295,490**	**Rp 440,000**

From the table it is known that there is a significant difference in the calculation of raw material inventory by conventional calculation and the EOQ method where EOQ calculations can save ordering costs and usage costs. Ordering costs before being calculated with EOQ of Rp 2,564,000 decreased to Rp 440,000 after being calculated using EOQ there is a savings of 82.84%. While the usage cost before being calculated using EOQ was Rp 110,812,200 decreased to Rp 110,295,490 there was a saving of 0.47%.

5. CONCLUSION

a. Ordering rice raw materials according to conventional calculations made by the nutrition installation of the kumala siwi hospital in 2023 is 96 times, ordering egg ingredients 120 times. ordering carrot raw materials 108 times, the frequency of ordering carrot raw materials 180 times, the number of times ordering watermelon raw materials 180 times, the number of times ordering catfish raw materials 72 times.
b. The cost of storing rice raw materials is 36.42%, chicken eggs are 15.19%, carrots are 20.83%, melon is 6.01%, watermelon is 8.18%, catfish is 13.39% of the total storage cost of Rp 24,780,000.
c. Decrease in frequency Ordering rice to 31 times in one year and 77 kg each time. Ordering chicken eggs to 25 times a year and 39 kg each time. Decrease in carrot ordering frequency to 27 times in one year and 51 kg per order. Decrease in ordering frequency of melon to 8 times in one year and 46 kg each order. Decrease in ordering frequency of watermelon to 10 times in one year and 51 kg each order. Decrease in ordering frequency of catfish to 21 times in one year and 40 kg each order.
d. There is a decrease in ordering costs and usage costs before using EOQ and after using EOQ. There was a booking cost savings of 82.84%, namely the booking fee before being calculated with EOQ of Rp 2,564,000 decreased to Rp 440,000 after being calculated using EOQ. While the savings in usage costs of 0.47% before being calculated using EOQ

amounted to Rp 110,812,200 decreased to Rp 110,295,490 after being calculated using EOQ.

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